Week 3: Assignment 3 (Non Graded)

Assignment not submitted

Note : This assignment is only for practice purpose and it will not be counted towards the Final score.

1) The model obtained by applying linear regression on the identified subset of features may differ from the model obtained at the end of the process of identifying the subset during *1 point*

- Best-subset selection
- Forward stepwise selection
- Forward stagewise selection
- All of the above

2) Given the following 3D input data, identify the principal component. *1 point*

1 1 9
2 4 6
3 7 4
4 11 4
5 9 2

(Steps: center the data, calculate the sample covariance matrix, calculate the eigen vectors and eigen values, identify the principal component)

- 0.9138
  -0.1035
  0.3926
- -0.2617

0.5891
0.7645

○
-0.4205
-0.6223
0.2342

○
-0.3105
-0.8014
0.5112

3) For the data given in the previous question, find the transformed input along the first two principal components. ***1 point***

○
6.9935 0.3021
2.7451 -0.2727
-0.9921 -0.4548
-4.5081 0.0449
-4.2383 0.3805

○
6.9935 0.4003
2.7451 -0.3876
-0.9921 -0.4110
-4.5081 1.6837
-4.2383 -1.2853

○
0.4003 0.3021
-0.3876 -0.2727
-0.4110 -0.4548
1.6837 0.0449
-1.2853 0.3805

○
3.4894 7.2079
-0.7590 6.4200
-4.4962 6.3966
-8.0122 8.4913
-7.7424 5.5223

4) Suppose you are only allowed to use binary logistic classifiers to solve a multi-class classification problem. Given a training set with 2 classes, this classifier can learn a model, which can then be used to classify a new test point to one of the 2 classes in the training set. You are now given a 6 class problem along with its training set, and have to use more than one binary logistic classifier to solve the problem, as mentioned before. You propose the following scheme (also known as one vs one approach in ML terminology) - you will first train a binary logistic classifier for every pair of classes. Now, for a new test point, you will run it through each of these models, and the class which wins the maximum number of pairwise contests, is the predicted label for the test point. How many binary logistic classifiers will you need to solve the problem using your proposed scheme? ***1 point***

○ 25
○ 15

- 18
- 12

5) With respect to Linear Discriminant Analysis, which of the following is/are true. (Consider a two class case) ***1 point***

- [ ] When both the covariance matrices are spherical and equal, the decision boundary will be the perpendicular bisector of the line joining the means.
- [ ] When both the covariance matrices are spherical and equal, the decision boundary will be perpendicular to the line joining the means.
- [ ] When both the covariance matrices are spherical and equal and the priors $\pi_1 = \pi_2$ then the decision boundary will be perpendicular bisector of the line joining the means.

Check Answers and Submit